MIL-T-46114B
13 July 1978
SUPERSEDING
MIL-T-46114A
18 October 1967

MILITARY SPECIFICATION

TAPE, GLASS-SETTING: RUBBER, UNCURED

(FOR SEALING GLASS WINDOWS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of glass-setting tape, a composition of uncured rubber and cotton fiber, used for sealing glass in metal channels (see 6.1)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

FSC 9320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contraction officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D 119 - Rubber Insulating Tape
D 412 - Tension Testing of Vulcanized Rubber
D 573 - Accelerated Aging of Vulcanized Rubber by the Oven Method
D 2240 - Indentation Hardness of Rubber and Plastics by Means of a Durometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Material. The material shall be a compound of uncured rubber and cotton fiber.

3.2 Construction. The glass-setting tape shall consist of 70 ± 3 percent uncured rubber and 30 ± 3 percent finely dispersed cotton fiber (see 4.6.5) evenly distributed throughout (see 4.4.1).

3.3 Physical properties. The glass-setting tape, when tested in accordance with 4.5 and 4.6 shall comply with the room-temperature requirements and after aging requirements of table I.

Table I. Physical property requirements

Characteristic	Requirement	Test paragraph
Room temperature requirements		
Durometer hardness (Shore A)	65 ± 5	4.6.2
Tensile strength (minimum)	250 psi (1.72 MPa)	4.6.3
Elongation (percent)	15 min.	4.6.3
Swell (dimensional change from original) maximum	±5 percent	4.6.4
After heat aging requirements		
Durometer hardness (Shore A)	65 ± 5	4.6.1 and 4.6.2
Tensile strength (maximum)	400 psi (2.76 MPa)	4.6.1 and 4.6.3
Elongation (percent)	24 min.	4.6.1 and 4.6.3

3.4 Dimensions. The length of roll, and width and thickness of tape shall be as specified (see 4.4.1 and 6.2).

3.4.1 Tolerances. The tolerances on glass-setting tape ahall be as follows:

Length of roll	-1 percent of specified roll length
Width	± 1/16 inch
Thickness	± .004 inch

3.5 Rolls. The glass-setting tape shall be furnished in rolls. Each roll shall be evenly and neatly wound around its own center without telescoping and shall be suitably restrained from unwinding. The tape shall unroll evenly and uniformly and shall not stick together so as to cause injury to the surface. Each roll shall be one continuous strip.

3.6 Color. The color of the glass-setting tape shall be black.

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3.7 Texture. The texture of the glass-setting tape shall be uniform.

3.8 Storage. The glass-setting tape shall meet the requirements of this specification when stored in original containers at moderate temperature (not to exceed 100°F) for six months (see 4.6.5).

3.9 Workmanship. The workmanship shall be such as to produce a tape, free of cracks, tears, cuts, voids, loose fibers and any imperfections which would be detrimental to its application.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot of glass-setting tape shall consist of that quantity of material produced in one continuous operation from one batch of raw materials at one place of manufacture and offered for delivery at one time.

4.2.1 Unit of product. The unit of product for the glass-setting tape shall be one roll.

4.3 Sampling.

4.3.1 Sampling for examination of tape. Unless otherwise specified (see 6.2), a random sample of rolls of glass-setting tape shall be selected for examination in accordance with level S-4 of MIL-STD-105. All of these rolls shall be examined for the defects listed in table II for "Rolls (Wound and unwinding)." Unless otherwise specified for the defects listed in table II under "Rolls (Unwound)" and "Dimensions of rolls" the same rolls shall be used, but the number of rolls reduced to level S-1 of MIL-STD-105. For rolls (Unwound) of table I, ten consecutive feet from each roll selected shall be examined.

4.3.2 Sampling for examination for preparation and delivery. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Unless otherwise specified, the number of samples shall be in accordance with level S-1 of MIL-STD-105. In addition, shipping containers fully prepared for delivery shall be examined for closure defects.

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4.3.3 Sampling for tests. Unless otherwise specified (see 6.2), two rolls of glass-setting tape shall be selected from each lot for all required tests (see 4.5 and 4.6).

4.4 Examination.

4.4.1 Glass-setting tape. Sample units selected in accordance with 4.3.1 shall be examined for defects and at the acceptable quality levels shown in table II.

4.4.2 Preparation for delivery. Sample units selected in accordance with 4.3.2 shall be examined for defects and at the acceptable quality levels shown in table III.

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TABLE II. Classification of defects for glass setting tape

Categories	AQL percent	Defects	Method of inspection	Requirement paragraph
Major				
101	4.0	Rolls not evenly and neatly wound	Visual	3.5
102		Rolls telescoping	Visual	3.5
103		Rolls not suitably restrained from unwinding	Visual	3.5
104		Tape does not unroll evenly and uniformly	Unroll	3.5
105		Tape sticks together causing injury to surface	Visual	3.5
106		Roll not one continuous strip	Visual	3.5
107		Cotton fibers not finely dispersed and evenly distributed	Visual	3.2
108		Tape color not black	Visual	3.6
109		Texture not uniform	Visual	3.7
110		Any cracks, tears, cuts, voids, loose fibers or detrimental imperfections	Visual	3.9
111	2.5	Length not within tolerance of specified length	Measure	3.4.1
112		Width not within tolerance of specified width	Measure	3.4.1
113		Thickness not within tolerance of specified thickness	Measure	3.4.1

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TABLE III. Classification of defects for preparation for delivery

Categories	AQL percent	Defects	Method of inspection	Requirement paragraph
Major		UNIT PACKAGE		
114	1.5	Tape not packaged in paperboard box	Visual	5.1.1.1
115		Carton damaged or no centering device	Visual	5.1.1.1
116		Wrong type of box	Visual	5.1.1.1
		INTERMEDIATE PACKAGE		
117	1.5	Number of rolls not as specified	Count	5.1.1.2
118		Wrong size container	Visual	5.1.1.2
119		Wrong type of container	Visual	5.1.1.2
120		Container improperly closed	Visual	5.1.1.2
		PACKING (BOX OPEN)		
121	1.5	Improper case liner	Visual	5.2
122		Wrong type of box	Visual	5.2
123		Wrong size box	Visual	5.2
		PACKING (BOX CLOSED)		
124	1.5	Lack of or improper strapping	Visual	5.2
125		Improperly closed	Visual	5.2
126		Gross weight exceeded	Scale	5.2
127		Marking improper or unidentifiable	Visual	5.3

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4.5 Tests. Specimens of glass-setting tape selected as specified in 4.3.3 shall be tested as specified in table IV.

Table IV. Test procedure

Characteristic	Test paragraph	Number of determinations	1/ Report results as
As received condition			
Durometer hardness	4.6.2	2/ Med of 5	Unit
Tensile strength	4.6.3	3/ Med of 3 or 5	psi (MPa)
Elongation	4.6.3	3/ Med of 3 or 5	percent
Swell test	4.6.4	2 width	Average percent
		2 thickness	Average percent
After heat aging			
Durometer hardness	4.6.1 and 4.6.2	Med of 5	Unit
Tensile strength	4.6.1 and 4.6.3	3/ Med of 3 or 5	psi (MPa)
Elongation	4.6.1 and 4.6.3	3/ Med of 3 or 5	percent

1/ Test reports shall include all values on which results are based.
2/ For definition of median (see 6.3).
3/ If one or more of the 3 specimens do not meet the requirement, 5 specimens shall be tested.

4.6 Test methods.

4.6.1 Heat aging. Heat aging shall be determined in accordance with ASTM D 573. The aging period shall be 48 ± 1 hours at an oven temperature of 70° ± 1.1°C (158° ± 2°F). To determine conformance to 3.3 the durometer hardness, tensile strength and elongation shall be determined prior to and after heat aging in accordance with 4.6.2 and 4.6.3.

4.6.2 Durometer hardness. The durometer hardness shall be determined in accordance with ASTM D 2240.

4.6.3 Tensile strength and elongation. Tensile strength and elongation shall be determined in accordance with ASTM D 412 with the exceptions listed in section 16 of ASTM D 119.

4.6.4 Swell test.

4.6.4.1 Swell test specimens. Two specimens shall be tested for swelling from immersion. The specimens shall be 2 ± 1/16 inches long and the width and thickness of the tape.

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4.6.4.2 Swell test procedure. Specimens (see 4.6.4) shall be immersed in distilled water at 23° ± 1.1°C (73.4° ± 2°F) for 48 ± 1 hours. The width and thickness dimensions shall be made and recorded prior to and after immersion. The average results of the two specimens for width and the average results for thickness shall be used to determine conformance to 3.3.

4.6.5 Certification of construction and storage life. The contractor shall submit to the procuring activity a letter certifying that the glass-setting tape shall meet the construction requirements specified in 3.2 and the storage life requirements specified in 3.8. The letter shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of this agent's authority to bind his principal. The Government reserves the right to check test material submitted by the supplier under certification.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Unless otherwise specified (see 6.2) each roll of tape shall be packaged in a standard commercial paperboard box designed with a centering device to prevent the roll from shifting within the box.

5.1.1.2 Intermediate packaging. Unless otherwise specified (see 6.2), 12 unit packages shall be overpacked in a fiberboard container conforming to PPP-B-636. Closure of the container shall conform to the appendix of the container specification.

5.1.2 Commercial packaging. Unless otherwise specified (see 6.2), the glass setting tape shall be packaged in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

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5.2.1 Level A. Unless otherwise specified (see 6.2), intermediate containers packaged as specified in 5.1 shall be packed in fiberboard boxes conforming to PPP-B-636, class weather resistant. When specified by the procuring activity, however, interior containers packaged as specified in 5.1 shall be packed in wood boxes conforming to the overseas type of either PPP-B-601 or PPP-B-621. Each exterior container shall be furnished with a sealed case liner conforming to MIL-L-10547 and appendix thereto. Case liners are not required however, when intermediate containers conform to PPP-B-636. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. The gross weight of fiberboard boxes shall be in accordance with PPP-B-636. The gross weight of wood boxes shall not exceed 150 pounds (68 kg).

5.2.2 Level B. Unless otherwise specified (see 6.2), intermediate containers packaged as specified (see 5.1) shall be packed in fiberboard boxes conforming to PPP-B-636 class domestic. When specified (see 6.2), intermediate containers shall be packed in domestic type wood boxes conforming to PPP-B-585, PPP-B-601 or PPP-B-621. Containers shall be closed and strapped in accordance with the applicable specification or appendix thereto. The gross weight of fiberboard boxes shall not exceed the weight limitation of the specification. The gross weight of wood boxes shall not exceed 150 pounds (68 kg).

5.2.3 Commercial packing. Unless otherwise specified (see 6.2) the glass setting tape shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The glass setting tape furnished under this specification is for use as a sealer for glass (windows) prior to insertion in metal channels (window lifts, vent windows, windshields).

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Length, width and thickness of tape (see 3.4).
c. If different sampling plan is required (see 4.3).
d. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).
e. Marking required (see 5.3).

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6.3 Median. If numerical values are arranged in ascending or descending order, the median is (1) the middle value of the series, if the number of values is odd and (2) the mean of the two middle values if the number of values is even.

6.3 Metric units. When metric dimensions are required, units for inch and foot may be converted to the metric equivalent by multiplying them by the following conversion factors:

English unit	Multiply by	Equals	Metric SI equivalent
inch	0.0254	=	metre (m)
foot	0.3048	=	metre (m)

Note: Conversion factors can be associated with ASTM E 380 entitled "Metric Practice Guide".

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